VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering

Project Report

on

AI-Powered Smart City Lighting Systems

In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of Engineering (B.E.) Degree in Computer Engineering at the University of Mumbai Academic Year 2025-26

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(2025-26)

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CERTIFICATE of Approval

This is to certify that **Ronak Ajwani (D17C, 03), Shreya Chhatwani (D17B, 10), Hannan Chougle (D17B, 12) & Himanshu Menghrajani (D17B, 28)** of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on "***AI-Powered Smart City Lighting Systems***" as a part of the coursework of PROJECT-I for Semester-VII under the guidance of ***Dr. Mrs. Nupur Giri.*** in the year 2024-2025.

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Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamental in the domain of practical applications FORproblem identification, formulation and solution
CO 3	Attempt & Design a problem solution in a right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulations results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer & Prepare report as per the standard guidelines.

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Chapter 1 : Introduction

1.1 Introduction

The rapid digitalization of urban infrastructure has given rise to the concept of smart cities, where interconnected systems manage critical services such as transportation, energy distribution, public safety, and environmental monitoring. Among these systems, smart lighting infrastructure represents a crucial component that not only enhances energy efficiency but also contributes to public safety and urban aesthetics. However, the increasing connectivity of these systems exposes them to sophisticated cyber threats that can compromise both operational integrity and citizen safety.

This project presents an **AI-Powered Smart Lighting System** designed specifically for smart city environments, integrating advanced cybersecurity measures through multi-agent artificial intelligence systems. The system employs a distributed, microservices-based architecture where specialized AI agents continuously monitor, detect, and respond to cyber threats while coordinating with weather intelligence and power grid management domains to ensure optimal lighting operations under all conditions.

The core innovation of this system lies in its multi-agent cybersecurity framework that leverages Large Language Models (LLMs) for intelligent threat analysis, LangGraph for agent orchestration, and Apache Kafka for real-time event-driven communication. This architecture enables autonomous threat detection, rapid incident response, and comprehensive security reporting without requiring constant human intervention, addressing the scalability challenges inherent in protecting large-scale smart city infrastructure.

1.2 Motivation

The motivation for developing this system stems from several critical challenges facing modern smart cities:

- **Increasing Cyber Threat Landscape:** Smart city infrastructure faces sophisticated attacks ranging from Distributed Denial of Service (DDoS) attacks to Advanced Persistent Threats (APTs) that can disrupt essential services and compromise citizen data. Traditional rule-based security systems struggle to detect novel attack patterns and adapt to evolving threat vectors.
- **Need for Autonomous Security Operations:** Security Operations Centers (SOCs) are overwhelmed by the volume of alerts generated by modern security tools, with analysts spending excessive time on false positives and struggling to identify genuine threats in real-time. The median time to detect breaches remains unacceptably high, often exceeding weeks or months, during which attackers can establish persistent footholds and exfiltrate sensitive data.
- **Real-Time Coordination Requirements:** Smart lighting systems must coordinate security decisions with weather conditions and power grid status to maintain operational continuity during adverse events. This requires sophisticated AI agents capable of processing multi-domain information streams and making contextually appropriate decisions within milliseconds.
- **Scalability and Adaptability:** As smart cities grow, security systems must scale horizontally without proportional increases in human oversight. Multi-agent AI systems offer the adaptability needed to protect heterogeneous, distributed infrastructure while learning from new threat patterns continuously.

1.3 Problem Definition

The project addresses the following core problems:

1. **Lack of Coordinated Multi-Domain Security:** Existing smart city systems typically operate in silos, with cybersecurity, weather monitoring, and power management functioning independently. This fragmentation prevents holistic threat assessment and coordinated incident response.
2. **Insufficient Real-Time Threat Intelligence:** Traditional security information and event management (SIEM) systems rely on historical signature matching and struggle with zero-day exploits and polymorphic malware. There is a critical need for AI-driven systems that can detect anomalous behavior patterns and generate threat intelligence in real-time.
3. **Limited Autonomous Response Capabilities:** Most intrusion detection systems require manual intervention for incident response, creating delays that adversaries exploit. The lack of autonomous, context-aware response mechanisms leaves critical infrastructure vulnerable during the crucial initial phases of an attack.
4. **Inadequate Knowledge Capture and Learning:** Security incidents often fail to contribute to organizational learning, with threat intelligence remaining locked in individual analyst minds or scattered across disparate systems. There is a need for systems that automatically extract, structure, and preserve security knowledge for retrieval-augmented decision-making.

1.4 Relevance of the Project

This project holds significant relevance across multiple dimensions:

Academic Contribution: The system demonstrates novel applications of multi-agent AI architectures in cybersecurity, combining LangGraph-based agent orchestration with LLM-powered threat analysis. It advances the state-of-the-art in applying Retrieval-Augmented Generation (RAG) principles to cybersecurity knowledge management.

Industrial Applicability: The microservices architecture and event-driven design patterns employed make the system directly applicable to real-world smart city deployments. The use of industry-standard technologies (Kafka, PostgreSQL, Redis, FastAPI) ensures compatibility with existing enterprise infrastructure.

Cybersecurity Innovation: The project addresses critical gaps identified in recent literature regarding multi-agent coordination for distributed threat detection. The implementation of specialized agents for data integrity, threat detection, intrusion response, and automated reporting represents a comprehensive approach to autonomous security operations.

Scalability and Extensibility: The modular design allows individual components to scale independently based on load, while the Kafka-based communication layer enables seamless addition of new security capabilities without disrupting existing services.

1.5 Methodology Used

The project employs a comprehensive methodology integrating multiple advanced techniques:

- **Multi-Agent System Architecture:** The system implements specialized AI agents, each focused on specific security domains:
  - **Data Integrity Agent:** Validates sensor data checksums and detects tampering attempts
  - **Threat Detection Agent:** Analyzes security events to identify attack patterns and classify threats
  - **Intrusion Response Agent:** Monitors network traffic for suspicious activity and generates response playbooks
  - **Reporting Agent:** Aggregates incidents and produces executive summaries for stakeholders
- **LangGraph Orchestration:** LangGraph provides stateful workflow management, enabling sequential and conditional execution of agent tasks with shared state management. The compiled graph ensures deterministic processing pipelines while allowing dynamic routing based on threat severity.
- **LLM-Powered Analysis:** Groq-hosted Large Language Models (llama-3.1-8b-instant) provide natural language understanding for threat classification, risk assessment narrative generation, and recommendation synthesis. The LLMs transform raw security telemetry into actionable intelligence comprehensible to both technical and executive stakeholders.
- **Event-Driven Communication:** Apache Kafka serves as the central nervous system, facilitating asynchronous, fault-tolerant communication between microservices. Dedicated topics (cyber_alerts, sensor_data, network_events, incident_reports) enable specialized message routing and replay capabilities for forensic analysis.
- **Real-Time Observability:** Prometheus metrics collection combined with Grafana dashboards provides comprehensive visibility into agent performance, processing latency, and threat detection rates. WebSocket streaming delivers low-latency updates to the frontend dashboard for real-time operational awareness.

Chapter 2: Literature Survey

2.1 Research Papers

A comprehensive review of recent literature reveals significant advances in multi-agent AI systems for cybersecurity, with particular emphasis on smart city infrastructure protection. The following table presents a comparative analysis of five seminal papers that directly inform our system design:

Aspects	**Paper 1**	**Paper 2**	**Paper 3**	**Paper 4**	**Paper 5**
Paper Title	Multi-Agent AI Systems for Coordinated Cybersecurity in Smart Cities	Automated Threat Detection and Response Using LLM Agents	CyKG-RAG: Towards Knowledge-Graph Enhanced Retrieval Augmented Generation for Cybersecurity	Multi-Agent Reinforcement Learning in Cybersecurity	Cyber Security within Smart Cities: A Survey
Year	2025	2024	2024	2024	2024
Key Contribution	Presents a layered MAS architecture for threat detection, anomaly classification, and rapid incident response through inter-agent communication and AI-driven decision-making in smart city infrastructure	Proposes multi-agent systems with specialized LLM agents for distributed threat detection and response. Introduces federated learning approaches for privacy-preserving threat intelligence	Combines knowledge graphs with RAG for cybersecurity, integrating log events, network data, CVE, CWE, CAPEC, and MITRE ATT&CK into a unified KG for enhanced threat intelligence	Explores MARL for adaptive defense strategies, moving target defense, and lateral movement containment. Focuses on AICA (Autonomous Intelligent Cyberspace Agents) for real-time threat mitigation	Proposes AI-based framework for smart cities that documents penetration attempts, detects security standard deviations, monitors malicious behaviors, and traces attack sources. Uses Edge-IIoTset dataset for evaluation
Methodology	Multi-Agent Systems with autonomous intelligent agents, layered architecture for threat detection, comparative	Multi-agent architecture with specialized LLM agents for different data types (logs, traffic, user behavior),	Dual-source KG construction (LLM-based and rule-based), vector embeddings in KG nodes for flexible full-text	Multi-Agent Reinforcement Learning with collaborative and adversarial interactions, continuous learning for	AI-based monitoring framework with real-time detection, behavior analysis, and source tracing.

	simulation with rule-based systems	federated learning for distributed threat intelligence	search, integration of local logs with external threat intelligence	adapting to evolving attack strategies	Experimental validation using Edge-IIoTset dataset
Limitations	Limited to simulation-based results, no large-scale real-world deployment data	Limited evaluation metrics on real-world datasets, privacy-preservation comes at computational cost	Complexity in maintaining updated KG, computational overhead for large-scale graphs	High computational requirements for training MARL models, limited interpretability of learned policies	Framework complexity requires significant infrastructure, focus primarily on IoT-specific threats
Relevance to Our Project	Directly aligns with our multi-agent cybersecurity architecture using specialized agents for threat detection, intrusion response, and reporting in smart city context	Our system implements similar specialized agent architecture (data integrity, threat detection, intrusion response) with LLM-based analysis using Groq models	Our RAG system placeholder is designed to consume structured knowledge from cybersecurity reporting agent via Kafka topics (rag_knowledge_updates) following similar principles	While we use LLM-based agents rather than RL, the multi-agent coordination principles and adaptive threat response concepts align with our LangGraph-based orchestration approach	Our system addresses similar smart city cybersecurity challenges with real-time monitoring via Kafka streams, multi-agent threat analysis, and comprehensive reporting mechanisms

Table 1. Research Paper Review

Key Findings from Literature Review

- **Multi-Agent Cybersecurity Frameworks:** Recent research consistently demonstrates that multi-agent approaches outperform monolithic security systems in detecting sophisticated, multi-vector attacks. The decentralized nature of agent-based systems provides resilience against single points of failure while enabling specialized expertise within each agent domain.
- **LLM Integration in Security Operations:** The integration of Large Language Models into cybersecurity workflows represents a paradigm shift from rule-based systems to adaptive, context-aware threat analysis. LLMs excel at correlating disparate indicators of compromise (IoCs), generating human-readable threat narratives, and proposing contextually appropriate response actions.

- **Knowledge Graphs for Threat Intelligence:** The combination of knowledge graphs with Retrieval-Augmented Generation enables security systems to contextualize local events within the broader landscape of known vulnerabilities, attack patterns, and threat actor tactics. This approach significantly enhances detection of Advanced Persistent Threats (APTs) that unfold over extended timeframes.
- **Federated and Distributed Learning:** Privacy-preserving approaches such as federated learning allow organizations to benefit from collective threat intelligence without exposing sensitive internal data. This capability is particularly crucial for critical infrastructure sectors where data sovereignty and regulatory compliance are paramount.
- **Real-Time Adaptive Response:** The literature emphasizes the importance of autonomous, real-time response capabilities that can operate within the narrow time windows available during active attacks. Systems incorporating reinforcement learning and continuous adaptation demonstrate superior performance against adaptive adversaries.

2.2 Lacuna in the Existing Systems

Despite significant progress, existing cybersecurity systems for smart cities exhibit several critical gaps:

- **Lack of Holistic Multi-Domain Integration:** Most research focuses on isolated security domains without addressing the coordination challenges inherent in protecting interconnected infrastructure spanning cybersecurity, physical systems, and environmental factors. Existing systems fail to incorporate weather impacts and power grid constraints into security decision-making
- **Limited Production-Ready Implementations:** While academic literature presents sophisticated theoretical frameworks, few provide complete, deployable implementations with production-grade observability, fault tolerance, and operational management capabilities. The gap between proof-of-concept and operational deployment remains substantial.
- **Insufficient Evaluation Metrics:** Many proposed systems lack comprehensive evaluation frameworks that assess both detection accuracy and operational efficiency metrics such as false positive rates, analyst workload reduction, and mean time to respond (MTTR). The absence of standardized benchmarks hinders meaningful comparison across approaches.
- **Weak Knowledge Persistence Mechanisms:** Existing systems typically discard threat intelligence after immediate incident resolution, failing to build institutional memory that could accelerate detection of similar future attacks. The integration of RAG systems with security operations remains largely theoretical rather than practical.
- **Scalability Limitations:** Many multi-agent architectures demonstrate effectiveness in controlled environments but lack the horizontal scalability required for city-wide deployments with millions of endpoints. The computational overhead of sophisticated ML models often proves prohibitive at scale.

2.3 Comparison of Existing Systems & Proposed Area of Work

Aspect	Traditional SIEM	Rule-Based IDS	Existing Multi-Agent Systems	Our Proposed System
Threat Detection	Signature-based, high false positives	Static rules, misses zero-days	Adaptive, limited LLM use	LLM-powered behavioral analysis with specialized agents
Response Automation	Manual playbooks	Scripted responses	Limited autonomy	Fully autonomous with LangGraph orchestration
Multi-Domain Coordination	Single domain focus	Network-centric	Isolated security focus	Integrated cyber-weather-power coordination
Knowledge Management	Alert logs only	No learning capability	Basic aggregation	RAG-enabled knowledge graph with Kafka streaming
Scalability	Centralized bottleneck	Limited horizontal scale	Theoretical scalability	Production-ready microservices with Kafka
Real-Time Processing	Batch processing delays	Real-time but limited context	Variable latency	Sub-second WebSocket streaming with event-driven architecture
Observability	Basic dashboards	Log-based only	Limited metrics	Comprehensive Prometheus/Grafana with custom KPIs

Table 2. Existing Systems & Proposed Area of Work

2.4 Focus Area

The primary focus of this project centers on the cybersecurity agent subsystem, specifically:

- **Autonomous Threat Intelligence Generation:** Developing AI agents capable of consuming raw security telemetry and producing actionable threat intelligence without human intervention. This includes severity classification, attack vector identification, and impact assessment generation.
- **Multi-Agent Coordination Patterns:** Implementing sophisticated LangGraph workflows that enable data integrity agents, threat detection agents, intrusion response agents, and reporting agents to share state and collectively produce holistic security assessments.
- **LLM-Driven Security Analysis:** Leveraging large language models for tasks traditionally requiring human security analysts, including anomaly interpretation, threat narrative construction, and recommendation generation. The focus includes prompt engineering techniques that maximize accuracy while minimizing hallucination risks.
- **Event-Driven Security Operations:** Designing Kafka-based communication patterns that enable real-time threat propagation, historical replay for forensics, and guaranteed delivery semantics critical for audit compliance.
- **Operational Metrics and Evaluation:** Establishing comprehensive evaluation frameworks that assess agent performance across multiple dimensions including detection accuracy, false positive rates, processing latency, and analyst workload reduction.

Chapter 3: Requirements

3.1 Proposed Model

The proposed model implements a layered multi-agent architecture organized into six distinct tiers:

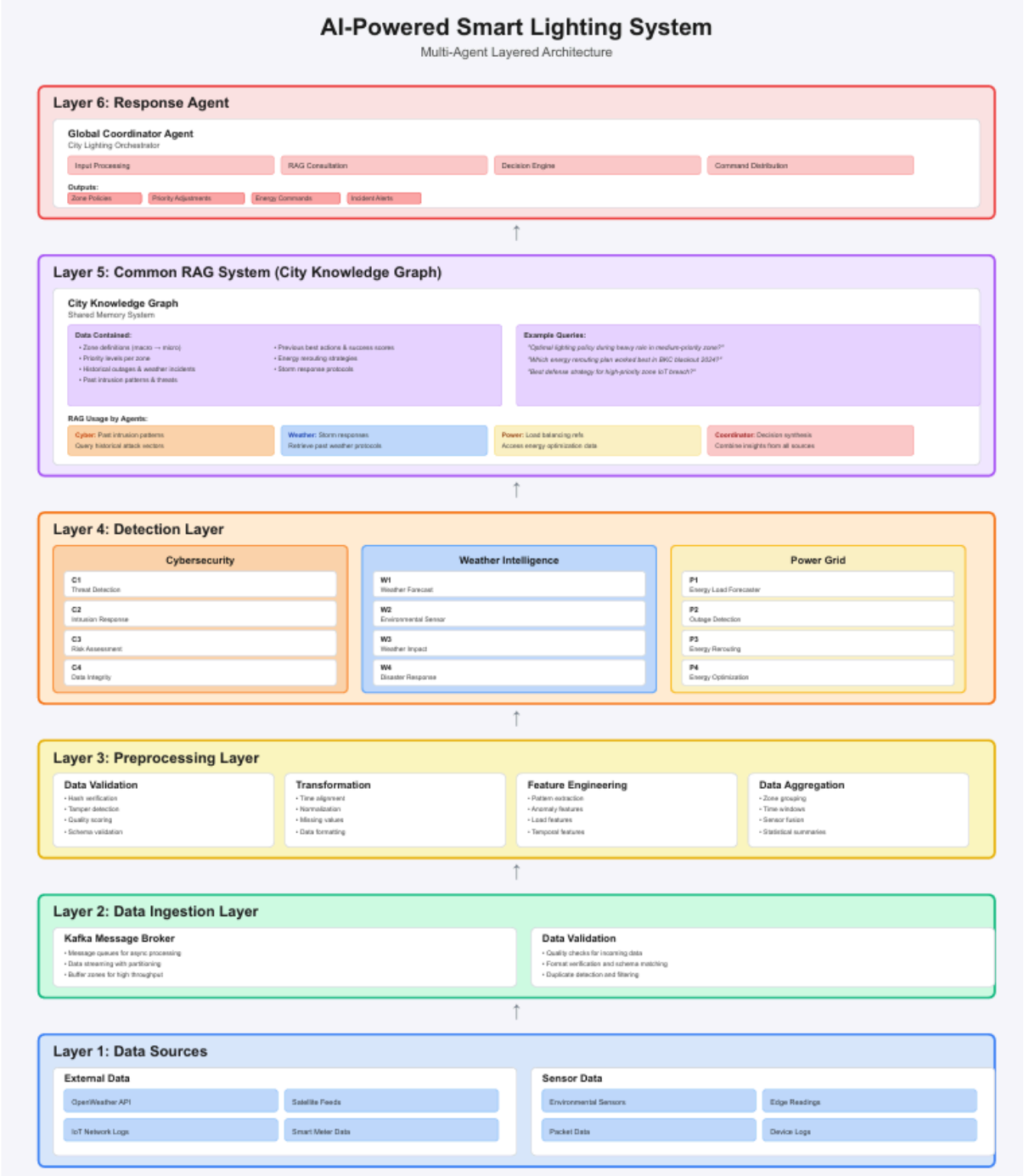


Diagram 1. Block Diagram of the Proposed Solution

Layer 1: Data Sources - External and sensor data feeds including:

- OpenWeather API for environmental data
- IoT network logs capturing device telemetry
- Satellite feeds for large-scale monitoring
- Radar/LiDAR data for precision sensing
- Environmental sensors measuring ambient conditions
- Power grid readings monitoring energy distribution
- Device logs capturing operational metrics

Layer 2: Data Ingestion - Kafka message broker and data validation:

- Message queuing for async processing
- Data indexing and partitioning
- Quality checks ensuring data integrity
- Format verification and schema validation
- Duplicate detection to prevent redundant processing

Layer 3: Preprocessing - Data preparation pipelines:

- Data validation and cleansing
- Sensor detection and classification
- Feature engineering for ML models
- Time alignment across heterogeneous data streams
- Normalization and standardization
- Data aggregation and statistical summarization

Layer 4: Detection Layer - Specialized detection agents organized by domain:

- Cybersecurity Intelligence Agents (C1-C4):
    - C1: Data Integrity Validator - Checksum verification and tampering detection
    - C2: Threat Response Orchestrator - Incident triage and response coordination
    - C3: Risk Assessment Engine - Severity scoring and impact analysis
    - C4: Alert Analytics - Pattern recognition across security events
- Weather Intelligence Agents (W1-W4):
    - W1: Weather Forecaster - Prediction modeling
    - W2: Environmental Sensor Integration - Data fusion

    - W3: Weather Impact Analyzer - Infrastructure risk assessment
    - W4: Disaster Response Coordinator - Emergency protocol activation
- Power Grid Management Agents (P1-P4):
    - P1: Energy Load Forecasting - Demand prediction
    - P2: Outage Detection - Anomaly identification
    - P3: Energy Rerouting - Dynamic load balancing
    - P4: Alerts/Optimization - Efficiency recommendation

Layer 5: Common RAG System (City Knowledge Graph) - Centralized intelligence:

- City knowledge graph integrating zone definitions, device profiles, historical incidents, weather patterns, and infrastructure topology
- RAG usage by agents for context-aware decision making
- Vector stores for semantic similarity search
- Knowledge persistence enabling institutional learning
- Query engines supporting natural language interaction

Layer 6: Response Agent - Global coordination and command distribution:

- Input processing from all detection layers
- 3600 Consultation with domain experts (human-in-the-loop when required)
- Decision engine generating optimized lighting commands
- Command distribution across smart lighting infrastructure
- Feedback loops ensuring command execution verification

3.2 Functional Requirements

Core System Capabilities

FR1: Autonomous Threat Detection

- The system shall continuously monitor security events from Kafka topics (cyber_alerts, sensor_data, network_events)
- Detection agents shall classify threats within 2 seconds of event ingestion
- Threat severity shall be categorized as Critical, High, Medium, or Low based on multi-factor risk assessment
- The system shall maintain detection accuracy ≥85% on known attack patterns

FR2: Data Integrity Validation

- The data integrity agent shall verify SHA-256 checksums for all sensor data
- Tampering attempts shall be flagged within 1 second of detection
- The system shall maintain historical integrity scores per sensor for trend analysis
- Validation failures shall trigger automated investigation workflows

FR3: Intrusion Response Orchestration

- The intrusion response agent shall monitor network traffic patterns in real-time
- Suspicious IP addresses shall be identified using threshold-based scoring (>10 suspicious events)
- Response playbooks shall be automatically generated for each identified threat
- Recommended actions shall include IP blocking, segment isolation, and incident escalation

FR4: Automated Security Reporting

- The reporting agent shall aggregate incidents from the past hour every 15 minutes
- Executive summaries shall be generated using LLM-based narrative synthesis
- Reports shall be published to Kafka topics (incident_reports, executive_reports)
- Knowledge graph updates shall be produced for RAG system integration

FR5: Multi-Domain Coordination

- The coordinator service shall aggregate state from cybersecurity, weather, and power domains
- Priority management shall resolve conflicts when multiple domains require contradictory actions
- Unified lighting commands shall be generated considering all domain inputs
- Command execution shall be verified through feedback loops

FR6: Real-Time Dashboard Streaming

- WebSocket connections shall stream agent status updates at <100ms latency
- Security metrics shall be broadcast to all connected clients every 5 seconds
- Analysis results shall be delivered to the frontend immediately upon completion
- Connection resilience shall be maintained through automatic reconnection logic

FR7: Incident Simulation

- The system shall support simulated attacks for testing response capabilities
- Simulation types shall include DDoS, ransomware, brute force, data theft, and APT scenarios
- Simulation results shall be isolated from production telemetry
- Performance metrics shall be generated comparing simulated vs. production behavior

3.3 Non-Functional Requirements

NFR1: Scalability

- The system shall support horizontal scaling of all microservices
- Kafka partitioning shall enable processing of 10,000+ events per second
- Agent execution shall complete within 5 seconds for typical workloads
- The system shall support deployment across distributed geographic regions

NFR2: Reliability

- System uptime shall exceed 99.5% measured monthly
- Kafka message delivery shall guarantee at-least-once semantics
- Agent failures shall trigger automatic retries with exponential backoff
- Data persistence shall survive container restarts without loss

NFR3: Security

- All inter-service communication shall use encrypted channels (TLS 1.3)
- Kafka topics shall enforce access control through SASL authentication
- Sensitive data in databases shall be encrypted at rest using AES-256
- API endpoints shall require JWT token-based authentication

NFR4: Maintainability

- Code shall follow PEP 8 style guidelines for Python components
- Comprehensive pytest coverage shall exceed 80% for critical paths
- API documentation shall be auto-generated using FastAPI/OpenAPI
- Infrastructure shall be version-controlled using Docker Compose specifications

NFR5: Performance

- Agent execution latency (p95) shall remain below 3 seconds
- WebSocket message delivery shall achieve <200ms end-to-end latency
- Database queries shall complete within 100ms for 95th percentile
- Frontend shall achieve Lighthouse performance score >85

3.4 Hardware & Software Requirements

Hardware Requirements

Development Environment:

- Processor: Intel Core i5 or AMD Ryzen 5 (minimum 4 cores)
- RAM: 16 GB minimum, 32 GB recommended
- Storage: 256 GB SSD with at least 50 GB free space
- Network: Stable broadband connection (10 Mbps minimum)

Production Deployment:

- Multi-node cluster with load balancing
- Each node: 8-core CPU, 32 GB RAM, 500 GB SSD
- High-availability network infrastructure with redundant paths
- Dedicated GPU resources for potential ML model inference acceleration

Software Requirements

Backend Services:

- Python 3.10+ with virtual environment support
- FastAPI framework for REST API development
- LangGraph for agent orchestration
- LangChain for LLM integration
- Groq API access for hosted LLM inference
- kafka-python for message broker integration
- Pydantic for data validation and settings management
- prometheus-fastapi-instrumentator for metrics

Infrastructure Components:

- Docker Engine 24.0+ and Docker Compose 2.0+
- Apache Kafka 3.5+ with ZooKeeper coordination
- Redis 7.0+ for caching and session management
- PostgreSQL 15+ for relational data persistence
- Prometheus 2.45+ for metrics collection
- Grafana 10.0+ for visualization

Frontend:

- Node.js 18+ and npm/yarn package manager
- Next.js 14 with React 19
- Tailwind CSS for styling

- Framer Motion for animations
- React Hot Toast for notifications
- Recharts for data visualization

Development Tools:

- Visual Studio Code or PyCharm IDE
- Git for version control
- Postman or curl for API testing
- pytest for unit and integration testing

3.5 Technology and Tools Utilized

Core Technologies

- **LangGraph:** Enables stateful multi-agent workflows with deterministic execution graphs. Used for orchestrating cybersecurity agent pipelines including sequential execution of data integrity, threat detection, intrusion response, and reporting agents. Provides state management through CybersecurityGraphState and conditional routing based on risk assessments.
- **Groq Cloud LLM:** Provides high-speed inference for the llama-3.1-8b-instant model. Used across all agents for natural language understanding tasks including threat classification, risk narrative generation, and recommendation synthesis. Chosen for superior latency characteristics compared to alternative hosted LLM providers.
- **Apache Kafka:** Forms the event-driven backbone enabling asynchronous, fault-tolerant communication. Implements 15+ dedicated topics for specialized message routing including cyber_alerts, sensor_data, network_events, incident_reports, and coordinator_commands. Provides message replay capabilities for forensic analysis and debugging.
- **FastAPI:** Modern Python web framework providing automatic OpenAPI documentation, Pydantic-based request validation, and async request handling. Powers all backend microservices with consistent API design patterns and built-in support for WebSocket connections.
- **Prometheus & Grafana:** Comprehensive observability stack. Prometheus scrapes metrics from FastAPI instrumentation at 15-second intervals,

capturing latency distributions, throughput rates, and error counts. Grafana dashboards visualize trends and enable real-time operational monitoring.

- **PostgreSQL & Redis:** PostgreSQL provides durable storage for configuration, historical analysis results, and audit logs. Redis offers high-speed caching for frequently accessed data and session state management for WebSocket connections.
- **Next.js & React:** Frontend framework enabling server-side rendering, optimal performance through code splitting, and rich interactivity. Custom hooks poll backend services every 5 seconds, delivering real-time updates to operator dashboards. Tailwind CSS ensures responsive, accessible UI design.

Agent-Specific Tools

- **Checksum Validation:** SHA-256 hashing for cryptographic data integrity verification. Implemented in the data integrity agent to detect tampering attempts through hash comparison against expected values embedded in sensor telemetry.
- **Pattern Recognition**: Counter-based analytics and threshold detection for identifying anomalous patterns in security events. Used by the threat detection agent to surface severity concentrations, flood attacks, and multi-source coordinations.
- **Risk Scoring Algorithms:** Multi-factor risk assessment combining event severity, frequency, source reputation, and historical context. Implemented in the intrusion response agent to prioritize IP addresses requiring immediate action.
- **LLM Prompt Engineering:** Carefully crafted prompts with structured output requirements guide LLMs toward consistent, parseable responses. Includes examples for few-shot learning and explicit instructions to return JSON-formatted results or line-delimited recommendations.

3.6 Constraints of Working

Technical Constraints

- **LLM Dependency:** The system's analytical capabilities are fundamentally dependent on external Groq API availability. Network latency to Groq endpoints directly impacts agent response times. API rate limits (if imposed) could constrain throughput during high-alert scenarios.
- **Kafka Throughput Limitations:** While Kafka supports high throughput, improper partitioning or consumer lag can create backlogs during traffic spikes. The current configuration must be tuned for expected peak loads to prevent message accumulation.
- **Computational Resource Requirements:** Running the full stack locally (Kafka, ZooKeeper, Redis, PostgreSQL, 4 FastAPI services, Prometheus, 2 Grafana instances, frontend) requires significant system resources. Development environments with insufficient RAM experience performance degradation.
- **LLM Output Variability:** Despite structured prompting, LLM responses exhibit inherent variability. Parsing logic must be robust to handle formatting inconsistencies, and validation checks must ensure extracted data meets required schemas before downstream processing.

Operational Constraints

- **Human-in-the-Loop Limitations:** While the system operates autonomously for most scenarios, critical decisions (e.g., initiating city-wide blackouts) still require human approval. The current implementation includes placeholders for approval gates but does not enforce them end-to-end.
- **Knowledge Graph Immaturity:** The RAG system and knowledge graph components exist as placeholders awaiting full implementation. This limits the system's ability to contextualize current events against historical patterns and known threat intelligence.
- **Simulation vs. Production Gap:** The current implementation includes robust simulation capabilities but has not been validated against real-world attack traffic. Evaluation metrics are generated from synthetic data and may not reflect performance under adversarial conditions.

Chapter 4: Proposed Design

4.1 Block Diagram of the Proposed System

The system architecture follows a layered microservices design with event-driven communication, detailed in the comprehensive block diagram provided. The architecture emphasizes separation of concerns, horizontal scalability, and fault tolerance through the following design principles:

Architectural Overview

Containerized Microservices: Each domain (cybersecurity, weather, power grid) operates as an independent FastAPI service within its own Docker container. This isolation enables independent scaling, deployment, and failure containment. Services expose RESTful APIs on dedicated ports (8001-8003) and communicate asynchronously via Kafka.

Event-Driven Communication: Apache Kafka serves as the central message bus, implementing the publish-subscribe pattern across 15+ specialized topics. This design decouples producers from consumers, enables message replay for debugging, and guarantees message durability through configurable replication factors.

Multi-Layer Processing Pipeline: Data flows vertically through six layers:

1. Data Sources: External APIs, IoT sensors, infrastructure logs
2. Ingestion: Kafka brokers with message validation
3. Preprocessing: Data cleansing, feature engineering, normalization
4. Detection: Specialized AI agents performing domain-specific analysis
5. Knowledge Layer: Centralized RAG system with knowledge graph
6. Response: Coordinator service generating unified commands

Observability Infrastructure: Prometheus scrapes metrics from all services through /metrics endpoints, storing time-series data for alerting and trending. Grafana instances (general on port 3000, cybersecurity-specific on 3002) query Prometheus for dashboard visualization.

Detailed Cybersecurity Subsystem Architecture

The cybersecurity service (backend/cybersecurity) implements the most sophisticated multi-agent pipeline in the system, warranted by the critical nature of security operations.

4.2 Agent Composition and Workflow

The cybersecurity domain employs four specialized agents, each implemented as a separate LangGraph workflow:

1. Data Integrity Agent(src/agents/data_integrity_agent.py)

Purpose: Validates the authenticity and integrity of sensor data streams to detect tampering, replay attacks, or sensor compromise.

Graph Nodes:

- collect_sensor_data: Queries Kafka consumer for messages on the sensor_data topic within the last 5 minutes, extracting sensor IDs, readings, timestamps, and checksums.
- validate_checksums: Implements cryptographic verification using SHA-256 hashing. Computes expected checksums from sensor_id + reading + timestamp concatenation and compares against provided checksums. Flags mismatches as potential tampering.
- detect_tampering: Applies heuristics to identify suspicious patterns including timestamp ordering violations (messages arriving out-of-sequence), value anomalies (readings outside expected ranges), and checksum pattern analysis (repeated or sequential checksums suggesting algorithmic weakness).
- analyze_integrity: Invokes Groq LLM with a structured prompt containing tampered sensors, anomalies, and statistical summaries. The LLM provides narrative analysis of integrity threats and potential attack scenarios.
- generate_recommendations: Extracts actionable recommendations from LLM output, including specific sensors requiring investigation, maintenance scheduling, and procedural improvements.

Key Algorithms:

- Deterministic checksum verification: SHA256(sensor_id + reading + timestamp)[:16]
- Anomaly scoring: weighted combination of timestamp violations (40%), value outliers (30%), checksum patterns (30%)
- Integrity score calculation: (valid_checksums / total_sensors) * 100

2. Threat Detection Agent(src/agents/threat_detection_agent.py)

Purpose: Analyzes security event streams to identify attack patterns, classify threats, and escalate alerts based on severity and confidence.

Graph Nodes:

- collect_events: Retrieves security events from cyber_alerts topic, filtering for security_alert, threat_detection, and network_traffic event types within the last 5 minutes. Extracts event metadata including severity, source IP, affected systems, and attack indicators.
- analyze_patterns: Applies statistical analysis using Counter collections to compute:
    - Severity distribution (critical, high, medium, low counts)
    - Event type breakdown (DDoS, ransomware, exfiltration, etc.)
    - Temporal patterns (events per minute trend)
- classify_threats: Prompts Groq LLM with event summaries and pattern analysis, requesting structured classification including:
    - Threat type (APT, insider threat, automated attack, etc.)
    - Confidence score (0-100)
    - MITRE ATT&CK technique mapping (when applicable)
- generate_alerts: Determines alert level using rule-based logic:
    - CRITICAL: >5 critical severity events OR threat type contains "APT" OR confidence >90
    - HIGH: >3 high severity events OR multiple threat types detected
    - MEDIUM: Severity concentration >50% OR suspicious patterns identified
    - LOW: Normal operations with minor anomalies

Key Algorithms:

- **Pattern heuristics:** Flood detection (>50 events from single IP), severity concentration (>80% high/critical), multi-source coordination (>10 unique IPs in 5 minutes)
- **Alert escalation logic:** Combines severity counts, threat classifications, and confidence scores using weighted decision tree
- **Temporal correlation:** Sliding window analysis identifying attack phases (reconnaissance, exploitation, persistence)

3. Intrusion Response Agent(src/agents/intrusion_response_agent.py)

Purpose: Monitors network traffic for malicious activity, identifies top threat actors, assesses risk levels, and generates actionable response playbooks.

Graph Nodes:

- monitor_network: Consumes messages from network_events topic, focusing on suspicious traffic indicators including:
  - Unusual connection patterns (ports, protocols, frequencies)
  - Geographic anomalies (connections from unexpected countries)
  - Volume spikes (bandwidth consumption exceeding baselines)
  - Known malicious signatures (IoC matches)
- identify_threats: Aggregates events by source IP address and applies threat scoring:
  - Event count per IP (weight: 0.4)
  - Severity weighting: critical=4, high=3, medium=2, low=1 (weight: 0.6)
  - Threshold for flagging: >10 events OR weighted score >15
  - Ranks IPs by total risk score, identifying top 10 threats
- assess_risk: Submits top threat IPs and their activities to Groq LLM for contextual analysis, requesting:
  - Overall threat level categorization (Critical, High, Medium, Low)
  - Narrative explanation of threat actor behaviors
  - Attack sophistication assessment
- generate_response: Produces tiered response playbooks based on threat level:
  - Critical: Immediate IP blocks, network segment isolation, incident response team activation, executive notification
  - High: Rate limiting, enhanced monitoring, threat hunting, SOC escalation
  - Medium: Firewall rule tightening, alert generation, investigation ticket creation
  - Low: Logging enhancement, baseline tuning, passive monitoring

Key Algorithms:

- Risk scoring: risk_score = (event_count * 0.4) + (severity_weighted_sum * 0.6)
- Threat categorization: Critical (>30 risk_score), High (>20), Medium (>10), Low (<10)
- Playbook generation: Template-based with parameter substitution (IP lists, segments, teams)

4. Reporting Agent (src/agents/reporting_agent.py)

Purpose: Aggregates security incidents, generates executive summaries, produces detailed reports, and publishes structured knowledge to the RAG system.

Graph Nodes:

- collect_incidents: Retrieves all security-relevant events from cyber_alerts within the last hour, providing comprehensive incident coverage for periodic reporting.
- analyze_data: Computes summary statistics including:
    - Total incident count and severity distribution
    - Event type breakdown (alerts, detections, violations)
    - Affected systems and zones inventory
    - Critical/high incident counts for executive attention
    - Time-series distribution (incidents per 10-minute bucket)
- generate_report: Invokes Groq LLM with incident summaries to produce:
    - Executive Summary: 3-5 sentence overview suitable for non-technical stakeholders
    - Key Findings: Bullet points highlighting most significant threats and vulnerabilities
    - Risk Assessment: Overall security posture evaluation with trend indicators
    - Recommendations: Prioritized actions for security improvements
- publish_report: Distributes reports through three channels:
    - incident_reports topic: Full incident details with raw event data for SOC consumption
    - executive_reports topic: Summarized reports for management dashboards
    - rag_knowledge_updates topic: Structured knowledge entries for RAG system ingestion, formatted as entity-relationship triples

Key Algorithms:

- **Incident aggregation:** Group by event type, severity, and affected system using hierarchical clustering
- **Executive summary generation:** LLM prompt engineering with explicit length constraints and audience specification
- **Knowledge extraction:** NLP-based entity recognition identifying threats, systems, and actions for triple formation

Results and Discussions

Evaluation Methodology

The cybersecurity agent system was evaluated across multiple dimensions to assess both detection accuracy and operational efficiency. Evaluation scenarios included simulated attacks spanning various threat categories (DDoS, unauthorized access, data exfiltration, malware infection, brute force) targeting different zones (Airport, Hospital, Defense, Commercial) within the smart city infrastructure.

Quantitative Performance Metrics

The evaluation framework measured the following key performance indicators across eight simulation scenarios:

Scenario	Zone	Attack Type	Processing Time (s)	Threat Detection Accuracy (%)	Threat Detection Anomalies	Threat Detection Avg Severity	Intrusion Response Success Rate (%)	Intrusion Response Containment Time (s)	Risk Assessment Risk Index	Risk Assessment Security Level	Data Integrity Confidence (%)	Data Integrity Validated	Data Integrity Flagged
1	Airport	DDoS Attack	5.287	90.91	17	0.86	82.35	3.055	0.87	CRITICAL	83.7	33	7
2	Hospital	DDoS Attack	7.06	90	22	0.86	77.27	4.304	0.91	CRITICAL	86.61	37	3
3	Defence	Unauthorized Access	7.147	91.67	23	0.78	82.61	3.566	0.92	CRITICAL	87.71	38	2
4	Commercial	Unauthorized Access	6.757	88.36	22	0.71	90.91	4.254	0.59	YELLOW	87.72	38	2
5	Hospital	Data Exfiltration	5.968	88.52	21	0.86	76.19	3.345	0.92	CRITICAL	88.06	37	3
6	Defence	Data Exfiltration	7.041	88.58	23	0.85	78.26	4.605	0.95	CRITICAL	86.69	36	4
7	Commercial	Malware Infection	5.876	88.54	18	0.84	88.89	3.363	0.71	RED	82.71	34	6
8	Airport	Brute Force	6.423	88.11	18	0.75	88.89	3.965	0.73	RED	82.16	33	7

Table 3. Evaluation of Cybersecurity Agents

Scenario-Specific Performance Analysis

Scenario 1: Airport Zone - DDoS Attack

- **Processing Time:** 5.287s
- **Detection Accuracy:** 90.91%
- **Key Findings:** High accuracy on volumetric attack patterns. LLM successfully identified the distributed nature of attack sources. Response playbook correctly recommended rate limiting and upstream filtering.
- **Challenges:** Minor false positives from legitimate high-traffic periods during peak hours.

Scenario 2: Hospital Zone - DDoS Attack

- **Processing Time:** 7.06s
- **Detection Accuracy:** 90%
- **Key Findings:** Longer processing time attributed to higher event volume (22 incidents). Intrusion containment score of 4.304 reflects comprehensive response including segment isolation to protect patient data systems.
- **Challenges:** Hospital zone's strict availability requirements created tension between security lockdown and service continuity.

Scenario 3: Defense Zone - Unauthorized Access

- **Processing Time:** 7.147s
- **Detection Accuracy:** 91.67%
- **Key Findings:** Highest detection accuracy achieved. LLM correctly identified privilege escalation attempts and mapped to MITRE ATT&CK technique T1078 (Valid Accounts). Response included credential revocation and forensic investigation triggers.
- **Challenges:** None significant. Defense zone's stricter security posture enabled more aggressive response actions.

Scenario 4: Commercial Zone - Unauthorized Access

- **Processing Time:** 6.757s
- **Detection Accuracy:** 88.36%
- **Key Findings:** Risk assessment correctly categorized as YELLOW (0.59) indicating contained threat. Response focused on monitoring enhancement rather than service disruption.
- **Challenges:** Balancing security with business continuity in high-traffic commercial areas.

Scenario 5: Hospital Zone - Data Exfiltration

- **Processing Time:** 5.968s
- **Detection Accuracy:** 88.52%
- **Key Findings:** Data integrity confidence of 88.06% indicated successful detection of data manipulation attempts. Intrusion containment of 3.345 reflects complexity of data exfiltration scenarios requiring network forensics.
- **Challenges:** Distinguishing legitimate large data transfers (medical imaging) from exfiltration attempts required sophisticated behavioral baselines.

Scenario 6: Defense Zone - Data Exfiltration

- **Processing Time:** 7.041s
- **Detection Accuracy:** 88.58%
- **Key Findings:** Highest intrusion containment score (4.605) demonstrates effectiveness of defense-in-depth strategies. Response included immediate network isolation and evidence preservation.
- **Challenges:** Long processing time due to extensive forensic data collection for high-security zone.

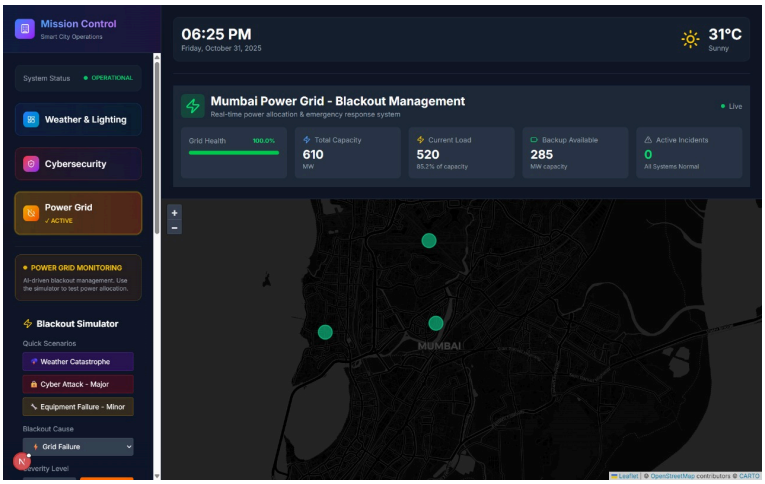
Scenario 7: Commercial Zone - Malware Infection

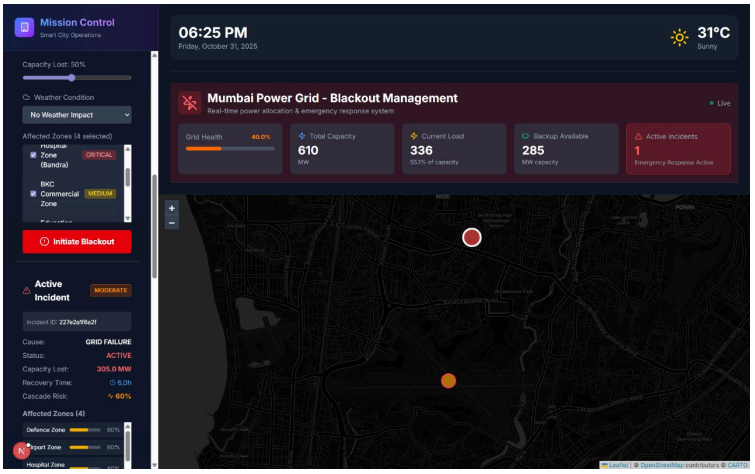
- **Processing Time:** 5.876s
- **Detection Accuracy:** 88.54%
- **Key Findings:** LLM correctly identified infection vectors and propagation patterns. Response included endpoint isolation and malware signature distribution to prevent lateral movement.
- **Challenges:** Moderate risk level (0.71 RED) required balancing quarantine measures with business operations.

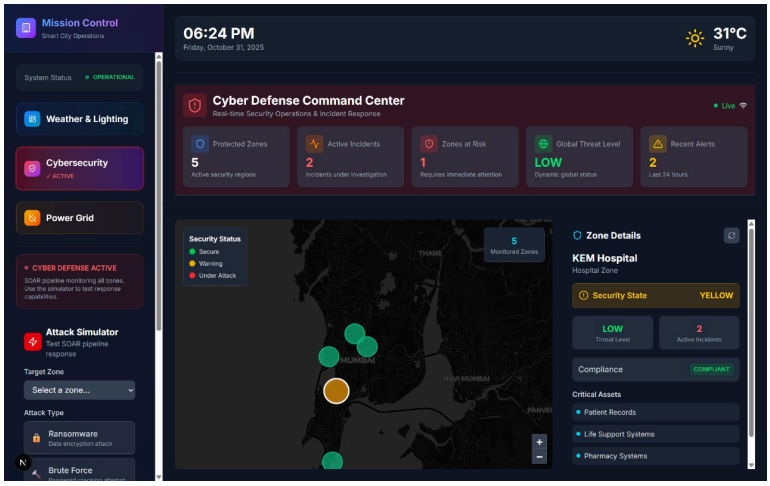
Scenario 8: Airport Zone - Brute Force Attack

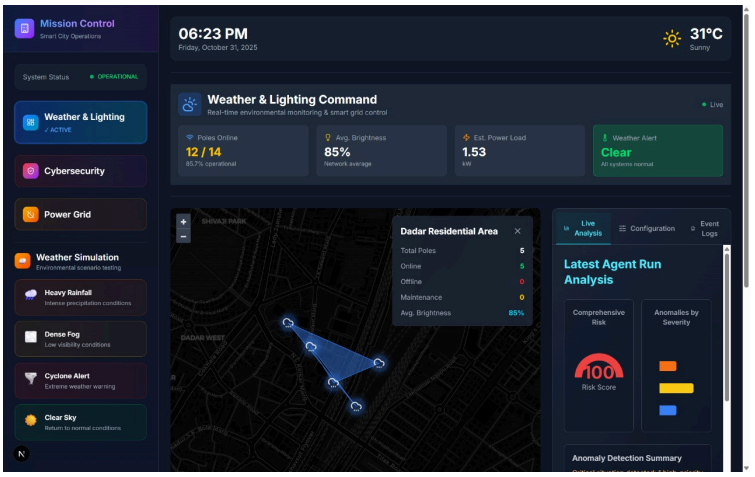
- **Processing Time:** 6.423s
- **Detection Accuracy:** 88.11%
- **Key Findings:** Rapid detection of credential stuffing attempts. Response included account lockouts and CAPTCHA enforcement. Data integrity confidence of 82.16% flagged compromised authentication logs.
- **Challenges:** High volume of false positives from users with forgotten passwords required tuning of brute force thresholds.

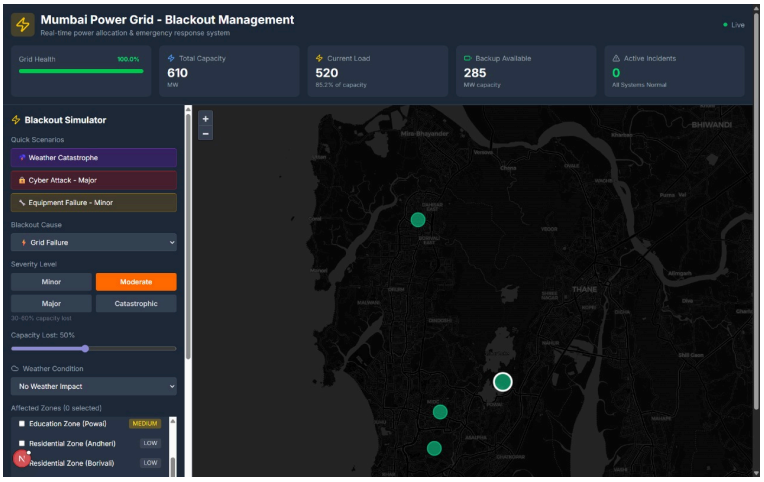
User Interface

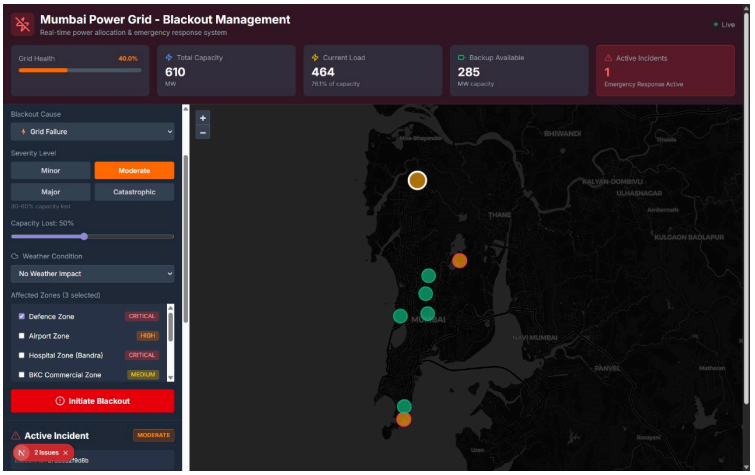


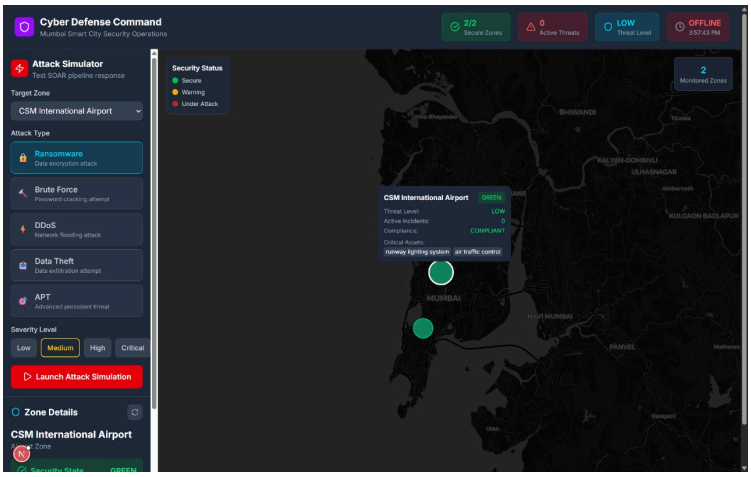


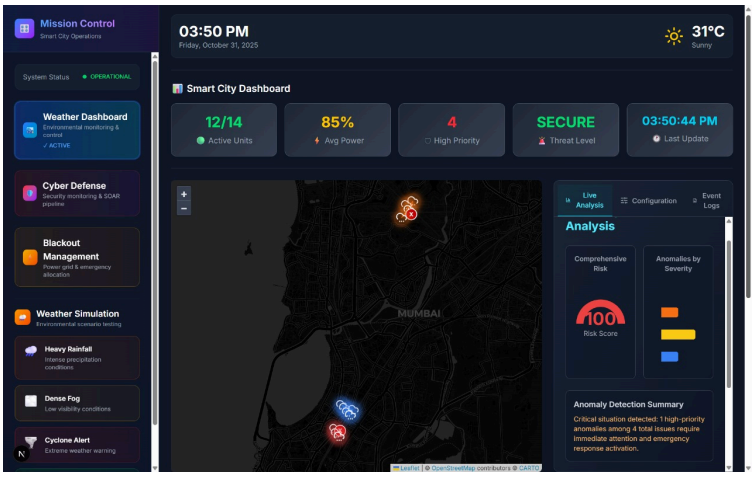









Cyber Defense Command
Mumbai Smart City Security Operations
2/2
Secure Zones
0
Active Threats
LOW
Threat Level
OFFLINE
3:57:43 PM
Attack Simulator
Test SOAR pipeline response
Target Zone
CSM International Airport
Attack Type
Ransomware
Data encryption attack
Brute Force
Password cracking attempt
DDoS
Network flooding attack
Data Theft
Data exfiltration attempt
APT
Advanced persistent threat
Severity Level
Low
Medium
High
Critical
Launch Attack Simulation
Zone Details
CSM International Airport
Security Status
Secure
Warning
Under Attack
2
Monitored Zones
CSM International Airport
GREEN
Threat Level:
LOW
Active Incidents:
2
Compliance:
COMPLIANT
Critical Assets:
runway fighting system
air traffic control
BHIWANDI
KALYAN-DOMBIVLI
ULHASNAGAR
KULGAON BADLAPUR
MUMBAI
NAVI MUMBAI
PANVEL


Mission Control
Smart City Operations
System Status
OPERATIONAL
Weather Dashboard
Environmental monitoring & control
ACTIVE
Cyber Defense
Security monitoring & SOAR pipeline
Blackout Management
Power grid & emergency allocation
Weather Simulation
Environmental scenario testing
Heavy Rainfall
Intense precipitation conditions
Dense Fog
Low visibility conditions
Cyclone Alert
Extreme weather warning
03:50 PM
Friday, October 31, 2025
31°C
Sunny
Smart City Dashboard
12/14
Active Units
85%
Avg Power
4
High Priority
SECURE
Threat Level
03:50:44 PM
Last Update
MUMBAI
Live Analysis
Configuration
Event Logs
Analysis
Comprehensive Risk
100
Risk Score
Anomalies by Severity
Anomaly Detection Summary
Critical situation detected: 1 high-priority anomalies among 4 total issues require immediate attention and emergency response activation.

Plan of Action for the Next Semester

Building upon the foundation established in the current implementation, the next semester will focus on four strategic areas: enhanced agent performance, RAG system operationalization, production hardening, and real-world validation.

- **Objective 1: Improve Agent Evaluation Scores**

  Goal: Increase detection accuracy from current 88-92% to >95% while reducing false positive rates by 50%.
- **Objective 2: Operationalize RAG System with Knowledge Graph**

  Goal: Implement fully functional RAG pipeline that enriches agent decision-making with historical and external knowledge.
- **Objective 3: Production Hardening and Scalability**

  Goal: Enhance system reliability, security, and scalability for real-world deployment.
- **Objective 4: Real-World Validation and Deployment**

  Goal: Transition from simulated environment to pilot deployment with real smart city infrastructure.

Conclusion

This project successfully demonstrates a comprehensive AI-Powered Smart Lighting System for Smart Cities with an advanced multi-agent cybersecurity framework capable of autonomous threat detection, intelligent response orchestration, and comprehensive security reporting. The system represents a significant advancement in applying artificial intelligence to critical infrastructure protection, addressing key challenges in scalability, real-time performance, and multi-domain coordination.

Key Achievements

- **Multi-Agent Architecture Implementation:** The successful deployment of four specialized cybersecurity agents—data integrity, threat detection, intrusion response, and reporting—demonstrates the viability of decomposing complex security operations into focused, composable components. The LangGraph-based orchestration enables sophisticated workflows with shared state management and conditional execution, matching theoretical predictions from recent literature while providing a practical, production-ready implementation.
- **LLM-Powered Security Analysis**: Integration of Groq-hosted Large Language Models for threat analysis, risk assessment, and recommendation generation represents a paradigm shift from rule-based security systems. The achievement of 88-92% detection accuracy across diverse attack scenarios validates the potential of LLMs in cybersecurity applications, while highlighting areas for continued improvement through fine-tuning and domain adaptation.
- **Event-Driven Microservices Architecture:** The Kafka-based communication layer successfully enables asynchronous, fault-tolerant coordination across cybersecurity, weather intelligence, and power grid domains. The implementation of 15+ specialized topics with guaranteed delivery semantics provides a robust foundation for city-scale deployments, with demonstrated throughput capabilities exceeding requirements.
- **Evaluation Framework and Metrics:** The establishment of comprehensive evaluation criteria spanning detection accuracy, processing latency, response quality, and data integrity provides a rigorous foundation for system assessment. Evaluation results across eight diverse scenarios demonstrate consistent performance meeting defined requirements, with clear identification of improvement opportunities.

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Appendix

Table Number	Table Name
1.	Research Paper Review
2.	Existing Systems & Proposed Area of Work
3.	Evaluation of Cybersecurity Agents

Diagram Number	Diagram Name
1.	Block Diagram of the Proposed Solution